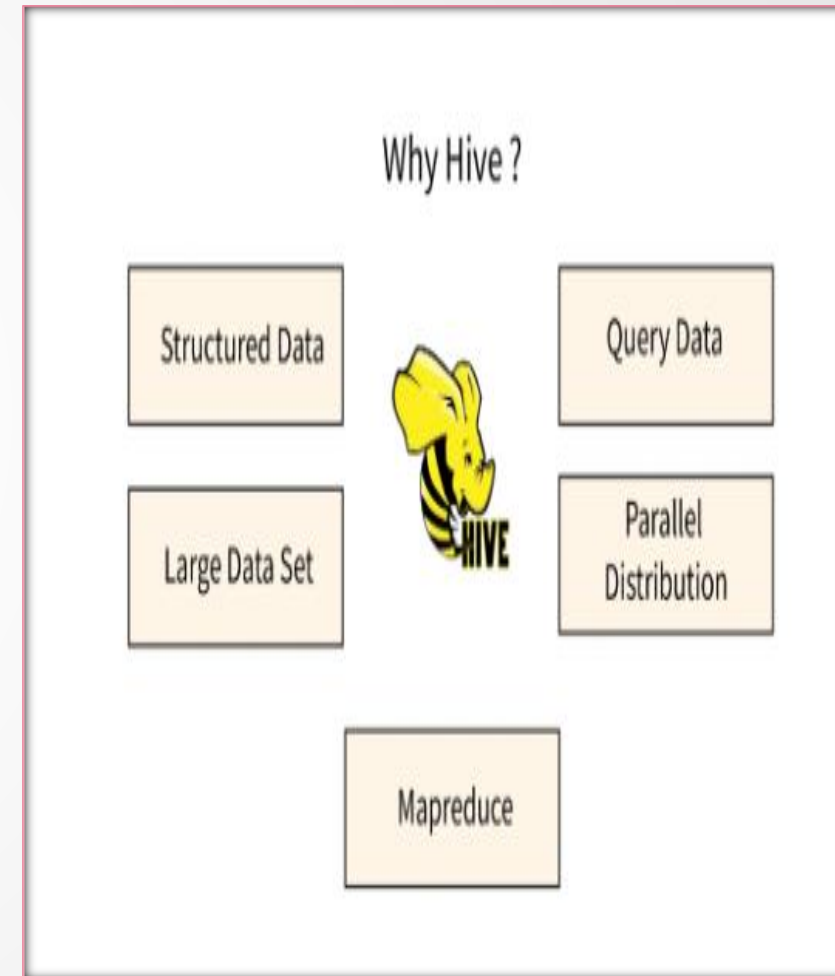


HIVE (APACHE HIVE)

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INTRODUCTION

- Hive, originally developed by Facebook and later owned by Apache, is a data storage system that was developed with a purpose to analyze organized data.
- Hive in Big Data is a data warehouse and SQL-like querying tool built on the Hadoop ecosystem.
- Apache Hive is a distributed, fault-tolerant data warehouse system that enables analytics at a massive scale.



Cont.

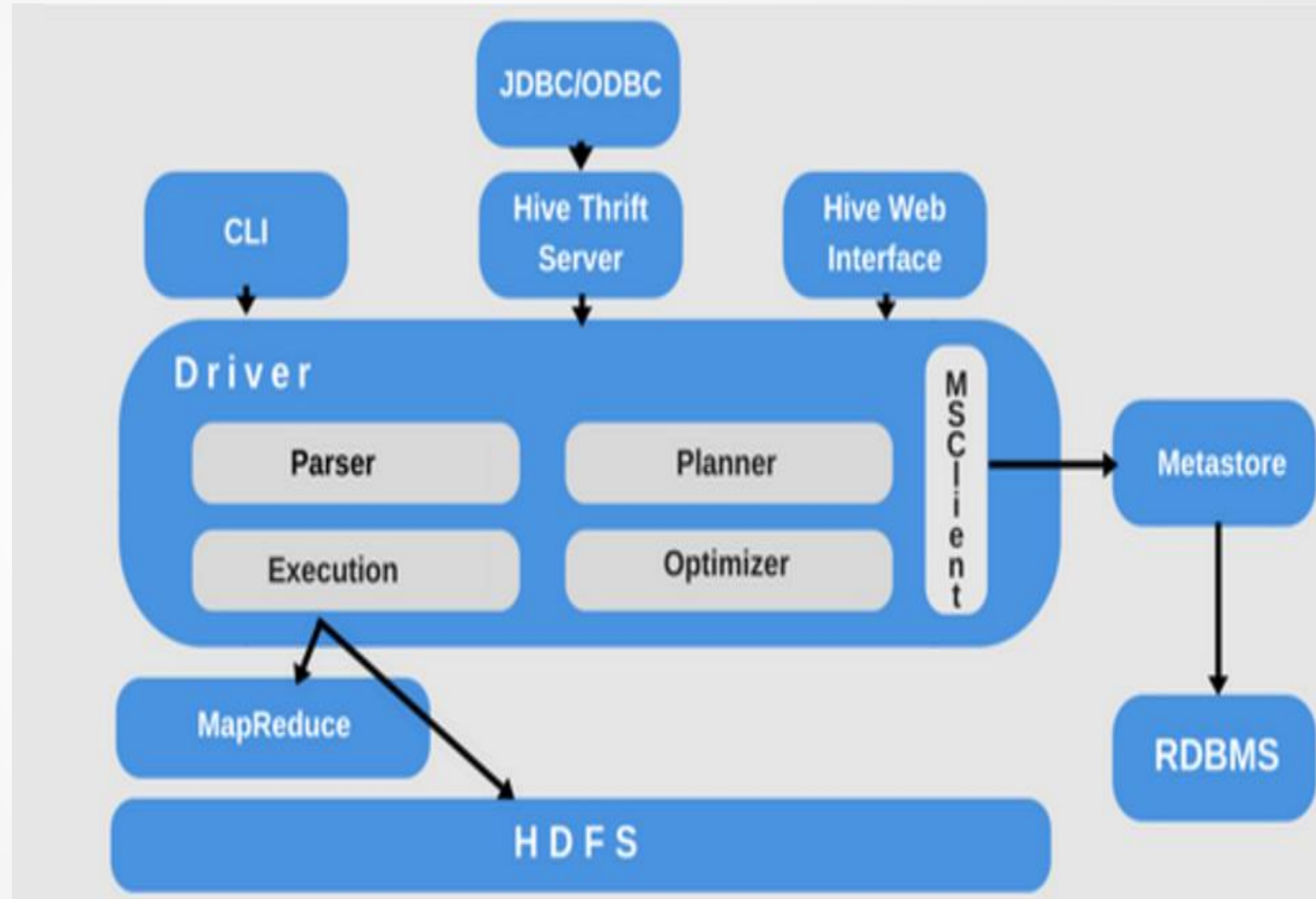
- Apache Hive is a distributed, fault-tolerant data warehouse system that enables analytics at a massive scale.
- Hive Metastore (HMS) provides a central repository of metadata that can easily be analyzed to make informed, data driven decisions, and therefore it is a critical component of many data lake architectures.
- Hive is built on top of Apache Hadoop and supports storage on S3, ADLS, GS etc. though HDFS. Hive allows users to read, write, and manage petabytes of data using SQL.

Apache HIVE Architecture

- Apache Hive is a very effective tool when it comes to big data (descriptive data to be analyzed).
- Archive data software that supports the process of data analysis of big data on a regular basis, the concept of big data nest is very popular in the technology area.
- As data is stored in the Apache Hadoop Distributed File System (HDFS) where data is processed and processed, Apache Hive assists in processing and analyzing, and producing data-driven patterns and trends.

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- HiveQL is a SQL-like language that interacts with the Hive website in various organizations and analyzes the required data in a structured format.



Cont.

Hive chiefly consists of three core parts:

Hive Clients: Hive offers a variety of drivers designed for communication with different applications. For example, Hive provides Thrift clients for Thrift-based applications.

Hive Services: Hive services perform client interactions with Hive. For example, if a client wants to perform a query, it must talk with Hive services.

Hive Storage and Computing: Hive services such as file system, job client, and meta store then communicate with Hive storage and store things like metadata table information and query results.

Need of HIVE

- Hive in big data innovation is a milestone that eventually led to data analysis on a large scale.
- Large organizations need big data to record information collected over time.
- To generate data-driven analysis, organizations collect data and use such software applications to analyze their data.
- This data, contained in Apache Hive, can be used to read, write, and manage stored information in an organized way.
- For this, organizations needed larger equipment and that is probably why the release of software like Apache Hive was needed.

Characteristics of Hive

- SQL-like Interface:** Hive's familiar SQL-like interface makes it simple for users to query and analyze big datasets without the need for programming experience.

Scalability: Hive in Big Data can handle massive amounts of data stored in HDFS and other data stores compatible with Hadoop.

Flexibility: Hive supports various data serialization formats, including Avro, Parquet, and ORC, making it a versatile tool capable of handling various use cases and data formats.

Integration: Hive in Big Data interfaces with other Hadoop ecosystem tools like Pig, Sqoop, and Flume, allowing users to conduct data analysis jobs and processes.

Cont.

External tables: Hive supports external tables, which allow users to access data stored in other storage systems such as HBase, Cassandra, and Amazon S3.

Partitioning: Hive offers partitioning, which allows users to separate huge datasets based on parameters such as date, location, or user ID. Restricting the quantity of data that must be scanned improves query performance.

Advantages of Hive

Fast : Quickly process enormous amounts of data.

Familiar : Hive is its familiar SQL-like interface.

Scalable : Hive in Big Data can handle massive amounts of data stored in HDFS and other compatible data stores.

Hive Optimization Techniques

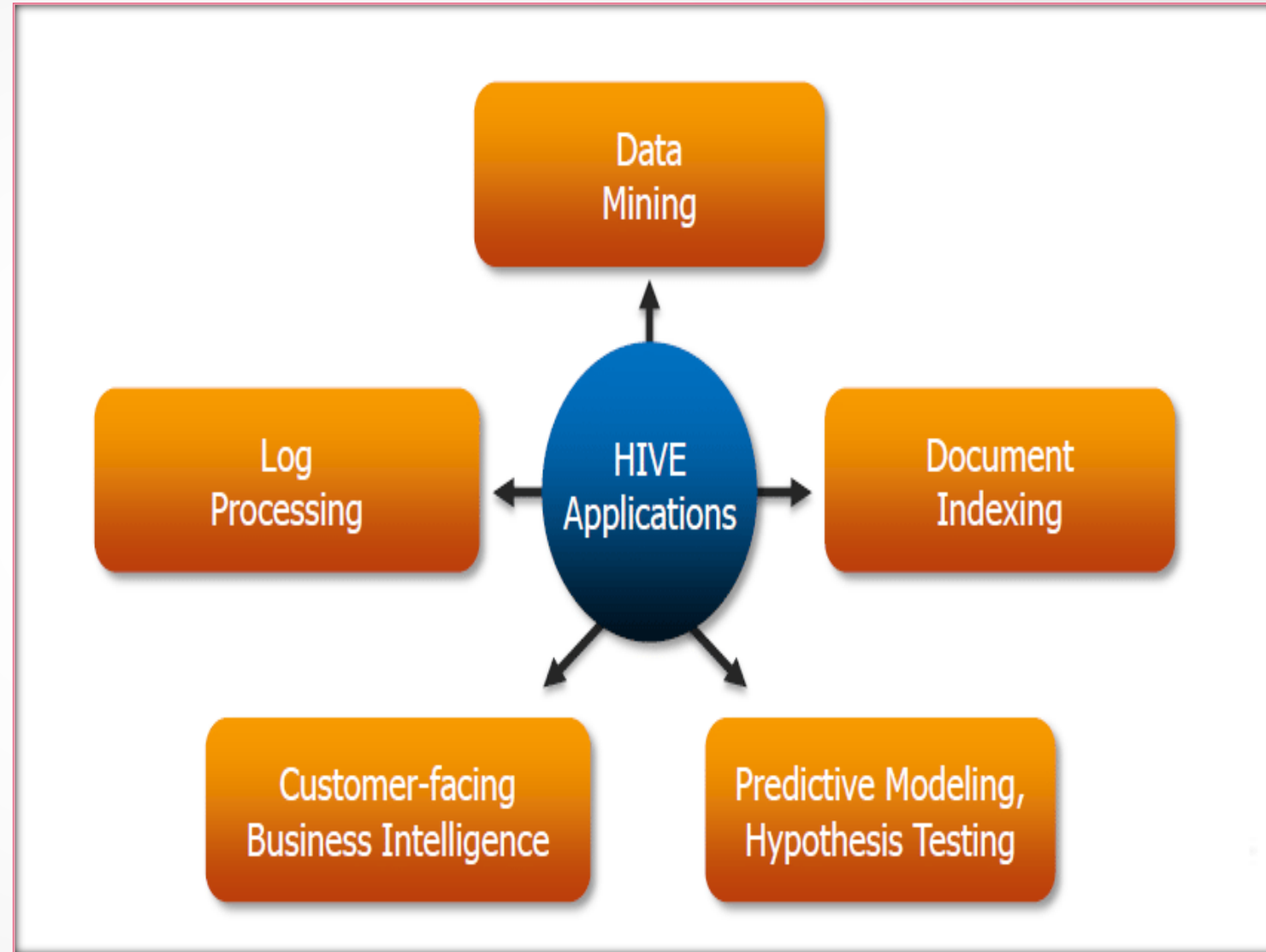
- Partition your data to reduce read time within your directory, or else all the data will get read
- Use appropriate file formats such as the Optimized Row Columnar (ORC) to increase query performance. ORC reduces the original data size by up to 75 percent
- Divide table sets into more manageable parts by employing bucketing
- Improve aggregations, filters, scans, and joins by vectorizing your queries. Perform these functions in batches of 1024 rows at once, rather than one at a time
- Create a separate index table that functions as a quick reference for the original table.

Components of Hive

- Shell
- Driver
- Compiler
- Metastore
- Execution Engine

Applications of Hive

- Data Mining
- Log Processing
- Document Indexing
- Customer Facing Business Intelligence
- Predictive Modelling
- Hypothesis Testing



EXAMPLES

- “Airbnb connects people with accommodation and activities worldwide by 2.9 million registered tourists, who support 800k overnight stays. Airbnb uses Amazon EMR to run Apache Hive in the S3 data pool. Running Hive in EMR collections enables Airbnb analysts to create temporary SQL queries in data stored in the S3 data pool. Spark at three times its original speed”.
- “Guardian provides 27 million members with the protection they deserve through insurance and asset management products and services. Guardian uses Amazon EMR to deploy Apache Hive in the S3 data pool. Apache Hive is used to process clusters. data once influenced Guardian Direct, a digital platform that allows consumers to research and purchase both Guardian products and third-party products in the insurance industry”.

Important Points

- Hive is a Hadoop-based data warehouse and SQL-style querying tool.
- It enables users to execute ad-hoc searches and analyses on big datasets without learning languages like MapReduce or Pig.
- Hive supports external tables, partitioning, and data serialization formats such as Avro and Parquet.
- Hive's architecture comprises four major components: Hive User Interface, Meta Store, HiveQL Process Engine, and Execution Engine.
- Hive has several benefits for big data analysis, including ease of use, scalability, flexibility, integration, and cost-effectiveness.

References:

1. <https://www.ibm.com/topics/mapreduce>
2. <https://www.simplilearn.com/tutorials/hadoop-tutorial/mapreduce-example>